Compound Degradations in Maritime Ship Recognition: a Paired Protocol, an Error-Union Decomposition, and Coverage of Co-occurrence as What Removes the Excess

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Abstract

Maritime images are rarely degraded by one factor at a time: haze, cloud and sun glare co-occur, yet maritime robustness studies vary one degradation at a time. We introduce a paired evaluation protocol for compound degradations, applied to fine-grained ship-type recognition on FGSC-23 (eight civilian classes): preprocessing is fixed, each image's corruption instance is a deterministic function of its identity, and every condition is evaluated per image, so any two models — or conditions — can be differenced without confounding. Four backbone families show compound conditions dropping accuracy by up to 43 points, non-linearly in the single-factor drops. In a pre-registered four-severity design, the compound drop is indistinguishable from its additive prediction at severities 2–4, but at severity 5 it exceeds that prediction by +11.3 pp (95% CI [+7.2, +15.3], BH q $<$ 0.001) and exceeds the largest single-degradation drop in 6 of 6 runs. Against the stronger baseline — the union of the constituents' error sets — 41.5% of severe-compound errors are made by no single degradation at any severity.

We then ask what removes the excess. Training on calibrated compound samples from six families reduces the excess on three pairs held out of every pool from +8.2 pp to −1.8 pp (clean accuracy 84.95% → 86.25%); a control showing the model the same six families one at a time, at the same rate and budget, stays at +7.2 pp (paired-minus-singles −5.8 pp, 6 of 6 cells). An arm matched on per-image damage recovers about 60% of the gain, so the accounting is roughly 60% under-matched damage and 40% co-occurrence; the co-occurrence part survives matching (−3.7 pp, 6 of 6) and is what transfers to unseen pairs. Neither part transfers to families absent from all pools, as pre-registered. Eight single-point interventions (seven mechanisms plus an initialization control) do not beat a strong combined-augmentation baseline, and we document one artefact that matters beyond this task: a +1.66 pp gain at a 24-epoch screening budget becomes −0.18 pp at 40 epochs. We release the protocol, harness, checkpoints and a manifest that reproduces every number.

Keywords: maritime computer vision; ship classification; robustness evaluation; compound degradations; superadditivity; negative results

1. Introduction

A ship photographed from shore in the Yellow Sea is not degraded by either haze or glare; it is degraded by some mixture of both, on top of an unknown amount of cloud. Recognition models deployed on such imagery are therefore evaluated under conditions that no single-corruption benchmark represents.

The published maritime literature has moved in the opposite direction. FoggySea isolates sea fog [13], CARP isolates cloud occlusion [14], and studies of haze on warship classification isolate haze. Benchmarks that do consider multiple corruptions — ImageNet-C [4] and its descendants — hold the number of types at one and vary only severity. Recent general-vision work studies composition as "two or more corruptions applied in sequence", without decomposing the joint effect against the marginals of its parts. As far as we can establish, no maritime study reports how a compound degradation compares to its constituents, and no study in any domain reports an explicit factorial decomposition of accuracy loss on ship-type recognition.

We therefore ask a narrower question than "how robust are ship classifiers": what does a compound degradation do that its constituents do not, and how does that difference depend on severity?

Answering it required three design decisions that turned out to matter more than any model:

1. Paired corruption. The corruption parameters for an image are drawn from stable_seed(recipe_version, sample_id, role, kind,

severity, domain). Two models evaluated on the same condition see the identical corrupted image; two conditions evaluated on the same model differ only in the degradation. This turns every model comparison into a paired comparison and removes most of the seed noise that otherwise dominates at 157 validation images.

2. Pre-registered screening with full-budget confirmation. Screening statistics were fixed before training. The protocol's value is shown by the fact that it caught its own artefact: a candidate that looked like +1.66 pp at a 24-epoch budget was −0.18 pp at 40 epochs.
3. Analysis of stored predictions, not just aggregates. Every condition's per-image prediction is archived, which lets us ask questions that aggregate accuracies cannot answer — in particular whether compound failures are the same failures a single degradation makes.

Contributions.

- A paired protocol for compound-degradation evaluation in maritime recognition, with a strong combined-augmentation baseline, group-disjoint folds and cluster bootstrap inference (§3).
- Evidence that the effect of compound degradation is not separable into its constituents, and that under the shipped renderer this non-separability appears as a threshold confined to the highest severity rather than a smooth gradient (§4.3) — including one pre-registered prediction that failed. A second, physically-motivated renderer, applied both to a frozen model and to models retrained under it, reproduces the severity-5 excess (10.2 / 16.0 pp against 9.7 / 12.6 pp) but produces it from severity 4 upwards: the excess is not an artefact of one rendering, while the onset severity is (§8.4).
- Evidence that compound failures are qualitatively new: 41.5% of severe-compound errors occur on images that every single degradation leaves correct (§4.4). Against the stronger independent-failure baseline (the union of single-corruption error sets), the severity-5 haze+glare excess survives in 6/6 cells (+7.3 to +13.1 pp, mean +9.9), while the haze+cloud excess vanishes (mean −0.7 pp); at severities 2–4 both pairs are mildly sub-additive (§4.6).
- A composition study over fifteen calibrated pairs: at matched marginal damage the interaction decomposes additively into per-family main effects (R^2 0.42–0.90 across two backbones and twelve cells), no single-corruption geometric predictor recovers the ordering, and the per-family values themselves are backbone-dependent (§4.6).
- Evidence that the excess is what an uncovered operating point looks like, and that it is removable: a pool of calibrated compound samples cuts the held-out excess from +8.2 to −1.8 pp (6/6 cells); the same six families applied one at a time do not (−5.8 pp paired, 6/6); ∼40% of the gain survives an arm matched on per-image damage (−3.7 pp, 6/6); and the effect transfers to unseen pairs but not to unseen families (§5). The same measurement on CIFAR-10-C reproduces the excess against the union baseline (+3.6 pp) while the additive baseline reports the opposite sign (−7.8 pp), and covering it there removes 8.3 pp at no clean cost, though by family exposure rather than pairing (§4.7).
- A pre-registered negative result on eight interventions, and a documented case in which a limited training budget manufactures a gain (§6).

2. Related work

Maritime datasets and degradation studies. FGSC-23 [15] provides fine-grained ship classes; the civilian subset used here has eight. FoggySea [13] benchmarks detectors on synthetically fogged SeaShips and reports monotone accuracy decay with fog density. CARP [14] studies few-shot ship classification under cloud occlusion at controlled coverage, and a study of FGSC-23 under synthetic cloud [8] reports uncertainty behaviour. All of these vary one factor.

Corruption benchmarks. ImageNet-C [4] established 15 corruption types × 5 severities with the mean corruption error (mCE) as the summary statistic; by construction it never combines types. ImageNet-3DCC [5] renders physically motivated corruptions but still one at a time. Mintun et al. [10] showed that augmentation-derived robustness is corruption-specific and that mCE can be gamed by similarity — a warning we take seriously and address by never tuning augmentation to the evaluation corruptions.

Composition. Two-factor perturbation studies [1] report that a second factor sometimes helps; compositional robustness work [9] stacks 2–6 corruptions and argues invariance does not transfer; decision-boundary analyses [12] examine which combinations break robustness. None decomposes the joint effect against the product of marginals — the quantity that distinguishes "harder" from "different".

Negative results as method. DomainBed [2] and RDumb [11] established that careful, well-controlled benchmarks often find simple baselines unbeaten. We follow that tradition rather than claiming an algorithmic contribution we do not have.

3. Protocol

Data. FGSC-23 [15], civilian scope, eight classes. 618 training and 157 validation images. Three group-disjoint folds over the training set (412/206 per fold), split by source_group_id so that no vessel or scene appears in both sides. The 236 images in the dataset's test split are not used for any result in this paper and are disclosed as compromised as a blind set (§8.3).

Preprocessing. Frozen: ImageOps.contain(image, (224, 224), BICUBIC) → corruption → ImageOps.pad((224, 224), color=(124, 116, 104)) → ImageNet normalization (mean [.485, .456, .406], std [.229, .224, .225]). No augmentation touches evaluation images.

Corruptions. Three physical families — haze (atmospheric transmission with random airlight), cloud (soft spatial support with alpha blending), glare (Gaussian bloom with gain) — each at five severities, implemented in degradations.py and frozen with a recipe hash recorded in every run. Compound kinds apply both components with independent random fields. Two domains: matched (fields drawn at the same scale as the scene) and shifted (fields drawn at a different scale, a stricter domain-shift control). 43 conditions are evaluated per model: clean, 30 single, 12 compound, plus one placement diagnostic.

Training. ConvNeXt-T [7] (torchvision ImageNet-1K weights, 27.8M parameters), batch 32, 40 epochs, cosine schedule, and a combined augmentation recipe that samples clean in 25% of batches, single degradations in 50% and compounds in 25%, with severities drawn from the recipe's own distribution. Two seeds (17, 29) per fold.

Inference. Cluster bootstrap over source_group_id, 2000–4000 resamples, with the two seeds of an image kept in the same cluster. Multiplicity controlled by Benjamini–Hochberg where a family of tests is reported. All gains are computed per image, same fold, same seed, same corruption instance.

Total cost. 136 training runs, ≈3.5 GPU-hours on a single RTX 4090.

4. Results

4.1. Single degradations are unequal, and haze is nearly harmless

On the baseline arm, clean accuracy is 84.5%. At severity 5, matched haze costs 3.1 pp on average (1.9–4.4 across runs), matched cloud 3.9 pp (2.4–5.8), and matched glare 11.3 pp (9.2–17.5). This asymmetry matters for the interpretation of every compound result below: a "haze + glare" compound is in practice a strong degradation with a weak one added, and haze alone is nearly harmless at every severity we tested.

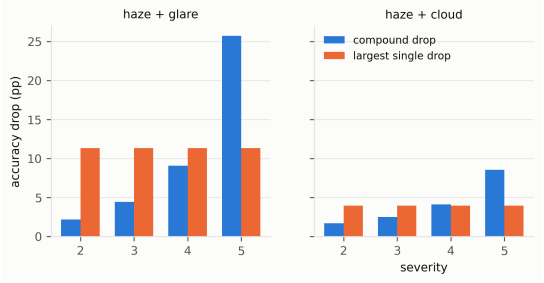


Figure 1. Accuracy drop (pp) of each compound condition against the largest drop produced by any severity of either constituent, baseline arm, mean over 3 folds × 2 seeds.

4.2. Compound degradations drop accuracy far below their constituents

At severity 5, haze+glare drops accuracy by 25.7 pp and haze+cloud by 8.6 pp, against largest single drops of 11.3 and 4.0 pp. Across four backbone families the same condition (matched_haze+glare_5) produces drops between −26.1 and −43.3 pp, and nine conditions have a mean drop of at least 5 pp with an interval excluding zero in all four. The compound effect is not a property of one architecture.

4.3. Superadditivity appears at severity 5 only — a threshold, not a gradient

We define superadditivity in accuracy space as

E(s) = [acc(clean) − acc(a+b, s)] − [acc(clean) − acc(a, s)] + [acc(clean) − acc(b, s)],

so that E(s) is the share of the compound drop not explained by the two marginal drops. The additive baseline is the sum of those two marginals, each measured on the same images, same model, same seed, at the same severity, in the same domain as the compound; E(s) is what the sum leaves unexplained. Because a reviewer can reasonably object that any such measure is an artefact of the metric, we report it in accuracy space (the margin-space version appears in the appendix and agrees in sign).

Severity	haze+glare E	95% CI	haze+cloud E	95% CI
2	0.00	[−1.45, +1.38]	−0.97	[−2.36, +0.41]
3	0.00	[−2.34, +2.20]	−0.49	[−2.20, +1.21]
4	+1.46	[−1.30, +4.24]	+1.21	[−0.98, +3.42]
5	+11.33	[+7.22, +15.22]	+1.62	[−0.97, +4.33]

Only haze+glare at severity 5 survives Benjamini–Hochberg correction over the eight estimates ($q < 0.001$); all others have $q \geq 0.48$.

We pre-registered four predictions before running severities 2 and 4. Three are confirmed and one failed; per-run values are in re-

ports/COMPOUND_ANALYSIS.md:

- P2 (E at severity 2 covers zero): confirmed.
- P3 (lower CI bound at severity 5 exceeds 2 pp): confirmed ([+7.22]).
- P4 (novelty share increases with severity): confirmed (3.8% → 6.6% → 19.1% → 41.5%).
- P1 (E increases monotonically with severity in at least five of six runs): failed — 2/6. The pooled mean is ordered, but the run-to-run noise (−3.4 to +4.9 pp at severities 2–4) exceeds the difference between adjacent severities.

We report P1's failure rather than redefining the test, because it changes the claim: the data do not support "superadditivity grows with severity". They support a sharper statement: at severities 2–4 the compound effect is additive, and at severity 5 it is not. In all six runs, E(5) exceeds every E(s ≤ 4) of the same run. With no point at severity 4.5, the crossover lies between 4 and 5 and cannot be localised further.

This statement is scoped to one rendering: under an independent, physically-motivated renderer with severity matched on transmission and saturated-pixel fraction, the severity-5 excess is preserved but its onset moves to severity 4. What survives is the excess and its growth with severity; the onset is a property of the renderer (§8.4, reports/PHYSICAL_RENDERER.md).

4.4. Compound failures are new failures

For each compound condition we ask, per image and per seed: is this image misclassified under the compound, and is it also misclassified under any severity of either constituent?

Condition	compound errors	error under no single at any severity	share
haze+glare, s=2	213	8	3.8%
haze+glare, s=3	241	16	6.6%
haze+glare, s=4	298	57	19.1%
haze+glare, s=5	504	209	41.5%
haze+cloud, s=2	207	2	1.0%
haze+cloud, s=3	217	12	5.5%
haze+cloud, s=4	237	17	7.2%
haze+cloud, s=5	292	67	22.9%

Error counts are per (fold, seed) evaluation, summed over the six runs; they share images across seeds and are descriptive, not independent observations. For the severity-3 and severity-5 conditions we additionally compute the overlap with the strongest single condition: 73.4% and 44.4% respectively for haze+glare (reports/COMPOUND_ANALYSIS.md).

At severity 5, four in ten compound errors are errors that no single degradation produces on that image — not even the strongest one, at any severity. Scope: it is bounded by the comparison set actually run — three families × five severities × two domains, the same model and seed. It does not claim that no conceivable single degradation could produce them. The phenomenon is not explained by the compound being simply more severe: there is no single degradation of comparable severity to compare against, since the compound drop (25.7 pp) exceeds the largest single drop (11.3 pp) in all six runs.

This is a behavioural analysis, not a mechanistic one: it does not say why the combination produces new errors, and the candidate mechanisms (feature loss at the boundary between two degradation fields, attention displacement by the glare point source under haze, class-boundary motion) are untested.

4.5. Where the excess comes from: two probes and their limits

The first probe measures the degradation pipeline, not the model. Over the 23 matched conditions, haze+glare at severity 5 saturates 60.2% of pixels (any channel ≥ 0.99) against 38% for the most saturated single condition (glare_5), and is the only condition combining high saturation with low contrast (luminance s.d. 0.094). Fitting the condition-level drop to saturation fraction and contrast over those 23 conditions gives $R^2 = 0.80$ (+30.7 / −16.7 pp); an interaction term raises it only to 0.814. Two boundaries apply: it is a condition-level relation, since per-image correlation between saturation fraction and misclassification is only 0.13 and 0.26 (two seeds); and it is incomplete, since haze+cloud at severity 5 saturates nothing, still exceeds the additive prediction and is the largest residual of the fit (≈6 pp).

The second probe asks whether hard highlight clipping — the mechanism the first probe points at — produces the excess. Everything is recomposed as corrupt() does, except that highlights roll off softly above 0.7 instead of clipping at 1.0, with the marginal drops re-measured in the same re-rendered world: the haze+glare_5 excess falls from +9.7 to +6.3 pp (saturation 60.2% → 40.7%). About a third of the excess is hard clipping; the rest survives without it. Single fold and seed, so directional rather than an estimate.

Turning the reading into a model change fails: a fourth input channel marking clipped pixels in any channel (zero-initialised mask stem, +1536 parameters), trained on the baseline recipe and scored on the same 43 paired conditions, moves compound accuracy by +0.61 pp ($p = 0.43$) and +0.97 pp ($p = 0.27$) on the two seeds, costs single-degradation accuracy (−0.47, −0.29 pp) and changes sign between seeds on the worst condition (−1.9 / +5.8 pp). Telling the network which pixels are censored does not help it.

Together the three probes support one statement only: saturation and contrast are a good description of the severity-5 drop at the level of conditions, and a poor one of which images fail. We do not claim a

mechanism for the new errors.

4.6. The interaction is additive over corruption families

Everything above concerns three hand-picked degradations. To ask whether the failure of additivity has structure beyond them, we added six synthetic families (defocus, motion, noise, rain, pixelate, impulse) with continuous levels, calibrated each level so that it alone costs a matched 10 or 20 pp on the frozen baseline, and measured all fifteen pairs over three folds × two seeds.

Raw E conflates interaction with error overlap: two degradations that break the same images cannot add up. Against the independent-failure baseline delta = e(a+b) − |E_a ∪ E_b|, the severity-5 haze+glare excess survives in all six cells (fold0/1/2 × seed17/29; +7.3 to +13.1 pp, mean +9.9), while haze+cloud falls to −0.7 pp and is positive in only two cells; at severities 2–4 both compounds sit 0.4–2.3 pp below the union baseline. The headline result is therefore not an artefact of overlap, but haze+cloud should not be called superadditive.

Across the fifteen added pairs, delta ranges from −4.4 to +18.3 pp (ConvNeXt-T). A per-family additive model delta(a,b) ≈ m_a + m_b explains R^2 = 0.48–0.90 in all six cells and 0.42–0.75 in all six ResNet-50 cells: which pair fails is largely predicted by which families it contains. The family values do not transfer between backbones (impulse +1.7 pp on ConvNeXt, −8.2 pp on ResNet-50), so superadditivity belongs to a (model, corruption) pair, not to a corruption pair alone.

No single-corruption geometric predictor recovers the ordering — mean feature-shift cosine, per-image displacement alignment, retained high-frequency energy and per-class error correlation all fail ($|\rho| < 0.6$ with unstable sign across cells). Predicting which composition will explode requires measuring part of the composition.

A six-cell replication gate fixed before folds 1–2 were run passed four of five criteria on ConvNeXt-T (the failed one, sub-additivity of two low-pass families, held in 4/6 cells); the ResNet-50 study passes all five but was not pre-registered.

4.7. The same measurement on a standard corruption benchmark

Everything above rests on one dataset and on corruption families written for this study. The decomposition is a measurement claim, so it has to be checked on a substrate other people use. We therefore repeat it on CIFAR-10-C [4]: the fifteen canonical corruptions at their five published severities, with a ResNet-18 [3] trained on clean CIFAR-10 [6] and then evaluated on 601 conditions — clean, 75 single and 525 pair — each on the full 10,000-image test set, with per-image correctness stored so that every decomposition can be recomputed offline. Nothing here is trained for robustness: the reference model sees clean images only, the setting the compound-corruption literature measures.

Two quantities are computed per condition. super = e(a+b) − e(a) − e(b) is the accuracy-space superadditivity compound-corruption studies report, built on additivity of marginal errors; delta = e(a+b) − |E_a ∪ E_b| is our union baseline, built on additivity of error sets. They differ by exactly the overlap, delta − super = |E_a ∩ E_b|, so the comparison is between two assumptions about what a second corruption does to an error already made: add a new error, or land on one that was going to happen anyway.

The archive's generator is unseeded, so infidelity is checked by comparing accuracy on regenerated against published corruptions: 74 of 75 cells agree within the pre-registered 1.0 pp tolerance (worst 1.17 pp, snow at severity 4). Five predictions (C1–C5) were frozen before the grid ran, with the specification that a failure of C1 — no positive excess at all — would demote the maritime result to a single-dataset observation.

Result. Over the 525 pair conditions the mean delta is +3.63 pp (median +2.84, 62% of cells positive, p10–p90 −9.5 … +18.8), inside the pre-registered [0, +8] band (C1). The same cells give a mean super of −7.76 pp: by the additive baseline the compound is easier than its parts, by the union baseline it is harder, and the gap between the two verdicts on identical data is the overlap — 11.4 pp of errors the additive baseline counts twice. Severity is not monotone (C5): delta climbs through +0.66, +2.93, +4.90, +6.58 and falls to +3.06 at s5, where the single corruptions already break most of the images the compound breaks. Pairs containing fog, snow or frost sit 4.2 pp above the pair median (C4: +7.02 against +2.84). One prediction fails: delta correlates only $\rho = +0.26$ with the damage the compound carries (C3, threshold 0.3), so the excess is not a law of how damaged the image is. Signs split by family: contrast (+10.9), fog (+10.5), motion blur (+9.4) and frost (+6.6) are superadditive, gaussian noise (−5.2) and impulse noise (−4.0) are subadditive, and the split does not follow from how strong each corruption is alone. The excess reproduces on a published benchmark; only its size changes, smaller here (+3.6 pp) than on the maritime data (+8.2 pp).

Does covering it transfer? Three arms apply §5's intervention here — pools of 15 and 40 pairs with five pairs held out of both, and a singles arm showing ev-

ery corruption alone at the same 0.5 rate — trained 200 epochs and evaluated on the same 601 conditions. The held-out excess at severity 3 falls from +7.86 pp (clean) to −0.17 (k=15) and −0.42 (k=40): monotone and 8.3 pp deep (B1), but k=15 to k=40 buys 0.25 pp, so the pool need not be wide, only complete. The pairing contrast is where the datasets part: singles reach +0.33 pp, 0.49 pp worse than the paired arm against a −1 pp threshold, so B2 fails, and a second seed repeats it (−0.89 pp), so the failure is not one draw. Clean accuracy holds (B3 , 94.3 / 94.2 / 95.0 vs 95.5). The effect transfers; the attribution does not: pairing beats family exposure by 5.8 pp on the maritime data and under 1 pp here across two seeds, so by B2's pre-registered consequence §5 claims family coverage.

5. Coverage: the excess is an operating point training never visits

The union decomposition of §4.6 says what the excess is not: compound images are not merely harder instances of existing errors. This section asks what removes it, under a deliberately unglamorous hypothesis — a model can only learn a joint operating point if training shows it one. Under the project's combined recipe, compounds are a quarter of the batches and are drawn from three families (haze, cloud, glare); the fifteen-pair lattice of §4.6 uses six families that training never mixes. If the excess is an uncovered operating point rather than an intrinsic property of compound degradation, teaching the model co-occurrence should remove it — and teaching it the same families one at a time should not.

Design. Six synthetic families (defocus, motion, noise, rain, pixelate, impulse), each calibrated to the same marginal cost: the level parameter is swept until a single family alone costs 20 pp on validation. Fifteen pairs arise from the six families; three (noise+pixelate, defocus+rain, motion+impulse) are held out of every training pool, so nothing below is evaluated on a pair the model could have memorised. Nested balanced pools c3, c6 and c12 contain 3, 6 and 12 distinct pairs over the remaining twelve. Every arm — including the re-measured baseline — gives each sample a clean view with probability 0.5 and a compound otherwise, so compound exposure is held constant and only the diversity of the compound distribution varies. Three folds × two seeds, 40 epochs, ConvNeXt-T, identical to §3.

Result. On the three held-out pairs the baseline excess is +8.2 pp (clean accuracy 84.95%). A pool of three pairs takes it to −1.8 pp (86.25), six pairs to +2.0 (86.08), twelve pairs to +1.4 (86.41). Two things separate here. First, the excess is removable: a handful of

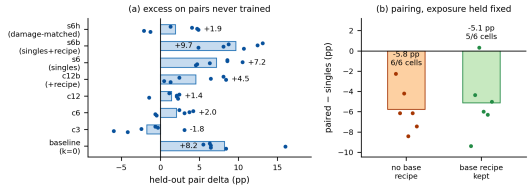


Figure 2. (a) Excess on the three held-out pairs for each arm, one dot per (fold, seed) cell; (b) the paired-minus-singles contrast, in which compound exposure is identical by construction, for the two recipe settings. Negative is better: it means the pair arm has less excess than the same families applied one at a time.

the right operating points turns an 8.2-pp excess into one of the opposite sign at no clean-accuracy cost. Second, our pre-registered dose-response prediction (P2: monotone improvement with pool size) failed: three pairs is as good as, or better than, twelve, by +3.5 pp (sd 2.2, 6 cells, range −0.5 to +5.3) on held-out pairs. What matters is not how many pairs the pool holds but whether each family is covered by at least one — every pool here contains all six, and extra pairs add nothing. Coverage of the family set, not pair diversity, is the operative variable.

arm	compound sampling	held-out delta (pp)	pool-pair delta (pp)	clean acc (%)
baseline (k=0)	recipe's own compounds	+8.2	+4.1	84.95
c3	3 pairs, all 6 families	−1.8	−0.7	86.25
c6	6 pairs, all 6 families	+2.0	+0.7	86.08
c12	12 pairs, all 6 families	+1.4	−0.2	86.41
s6	6 singles, 20 pp each	+7.2	+4.9	86.73
s6h	6 singles, 40 pp each	+1.9	+1.2	86.08
c12b	12 pairs + base recipe	+4.5	−0.1	85.84
s6b	6 singles + base recipe	+9.7	+5.9	86.33

Three folds × two seeds per arm; every arm applies a compound to half of every batch, so the compound rate is 0.5 everywhere. "Pool pairs" are the pairs a pool contains; the held-out pairs appear in no pool.

What the gain is not. Family exposure is the obvious alternative explanation, so s6 applies the same six families at the same 50/50 rate, one per draw instead of two: it lands at +7.2 pp (clean 86.73), almost exactly the baseline. Against c12 the paired-minus-singles difference is −5.8 pp (range −8.4 to −2.3), negative in 6 of 6 cells; on the twelve pool pairs −5.1 pp; clean accuracy differs by −0.3 pp. Adding the same information one family at a time does nothing; showing the model the co-occurrence does. The result also holds when the project's own combined recipe is kept in the mixture rather than replaced (§3's recipe, which trains haze, cloud and glare): c12b reaches +4.5 against s6b's +9.7, a paired-minus-singles difference of −5.1 pp (5/6 cells negative, pool pairs −6.0 pp, 6/6).

Damage per image is a real confound, and it is part of the effect. A compound image carries roughly twice the damage of a single-family image. We removed this in a third control: s6h applies one family per draw at a

level calibrated so that a single corruption costs 40 pp, the same damage per image a pair carries. It reaches +1.9 pp (clean 86.08). Our pre-registered prediction (P12: $\geq +3.0$ pp, i.e. that damage-matching would not reproduce the gain) is falsified: matching the damage recovers 5.3 of the 8.9 pp that separates s6 from c3. What survives is smaller but consistent — at equal damage per image, pairs still beat singles by -3.7 pp (range -7.1 to -1.8), 6 of 6 cells, on pairs seen in no training pool. The honest decomposition of the 8.9-pp gain is therefore roughly 60% damage per image and 40% co-occurrence, and the co-occurrence part is the part that survives matching.

Where it stops. Coverage removes the excess only for pairs drawn from families the pools cover. The paper's haze+glare condition is never in any lattice and its families are absent from the lattice recipe: with the base recipe retained (c12b vs s6b), its severity-5 accuracy is 56.1% against 57.4% and its excess 11.0 pp against 12.0 pp, no improvement, as the pre-registered negative prediction (P10) expected. Coverage transfers to unseen pairs over covered families; it does not transfer to unseen families. That is a boundary of the mechanism, not a defect of the experiment: the excess is what an uncovered operating point looks like, and covering the compositions at the families you expect to meet is the operative recipe.

6. Interventions: a pre-registered negative result

Seven interventions across three loci plus one initialization control, each trained at 40 epochs, three folds, two seeds, with identical data, preprocessing, corruption instances and evaluation. Screening criteria were fixed in advance: mean gain $\geq +1.5$ pp on the combined conditions, two of three folds positive, no clean-accuracy loss beyond 1 pp, and a paired bootstrap interval excluding zero.

Candidate	Locus	Mean Δ	95% CI	Verdict
T1 consistency training	training	−0.18	[−1.13, +0.76]	not sustained
I1 conditional normalization	input	+0.94	[−0.09, +1.98]	below criterion
R3 feature denoising	representation	−0.69	[−1.76, +0.35]	no positive evidence
R1 multi-scale fusion	representation	−1.10	[−2.23, +0.05]	interval covers zero
I2 Retinex preprocessing	input	−2.56	[−3.86, −1.22]	negative
T2 group DRO	training	−5.04	[−6.52, −3.54]	negative
T3 curriculum	training	−8.76	[−10.14, −7.40]	negative
T4 ImageNet-22K init	initialization	−0.71	[−2.07, +0.52]	no advantage demonstrated

Two caveats we insist on. First, R1's interval covers zero: the correct statement is "no detectable gain at +4% parameters and +8.5% latency", not "significant harm". Second, T4 compares two specific weight sets, not "pretraining quality" as a factor.

The budget artefact. T1 showed +1.66 pp [0.63, 2.68] when screened at 24 epochs and −0.18 pp [−1.13, 0.76] at 40 epochs. Both arms improved with the longer budget (84.5 → 85.7 and 84.0 → 85.1 macro-F1), and the candidate's apparent advantage was the difference between two convergence trajectories. Any screening pipeline that compares candidates at a truncated budget inherits this risk; our protocol's full-budget confirmation is what caught it.

Error-flow analysis. Across 6 (fold, seed) pairs × 8 conditions × 206 images, every candidate fixes some baseline errors and breaks more: R1 620 fixed / 728 broken, I2 639/904, T2 608/1103, T3 547/1415. These counts are not independent observations (the same image recurs across conditions and seeds) and are used only to describe direction; formal inference remains the clustered bootstrap.

Clean accuracy does not rank candidates. The curriculum arm had the best clean accuracy of any candidate (85.1%) and the worst compound accuracy (67.3%); the ImageNet-22K initialization gained +0.73 pp clean and lost −0.71 pp compound. Neither is significant alone, but together they argue against selecting models by clean validation accuracy when deployment is under weather.

7. Real imagery: observational only

On 157 FGSC-23 validation images annotated for real degradation (93 with at least one), the accuracy difference between degraded and clean groups is −2.3 pp, seed interval [−7.8, +4.3]; on 184 two-class FGSRCS development crops it is +1.8 pp. Neither can support a claim that real degradation hurts accuracy here — the heavy-degradation stratum holds at most two images — and we report them only to bound what we claim: the findings concern the synthetic protocol, not their transfer to real weather.

8. Discussion and limitations

8.1. What this work supports

- Under a fixed data set, recipe and pipeline, compound degradations are not additive at the highest severity: the joint drop exceeds the sum of marginals by 11.3 pp and exceeds the largest single drop in every run, and 41.5% of severe-compound errors come from no single degradation — ruling out "just harder" at the level of behaviour.
- The excess is not intrinsic to compound degradation. A pool over the same six families turns an +8.2 pp held-out excess into −1.8 pp at no clean-accuracy cost; the same families applied one at a time leave it at +7.2 pp (§5). About 40% of that gain survives damage matching, and only that part transfers to unseen pairs. On CIFAR-10-C single families recover the same 8.3 pp (§4.7).
- Seven mechanism-diverse single-point interventions and one initialization control do not beat a strong

combined-augmentation baseline. The negative result is pre-registered and reproducible.

8.2. What it does not support

- No performance ceiling. Frequency-domain methods, deeper backbones, larger data and multi-factor interventions were not tested.
- No mechanism. The novelty analysis is behavioural. The saturation/contrast fit of §4.5 is a condition-level association ($R^2 = 0.80$) with weak per-image correlation ($r = 0.13, 0.26$) and is not evidence of per-image causation. Both mechanism tests derived from it are negative or partial: removing hard clipping removes about a third of the excess, and a clipped-pixel mask gives no gain. A per-image drift signal predicts failure (AUC 0.78–0.82), but as a training weight it fails in both directions ($\alpha = +1$: -1.29 and $+0.16$ pp; $\alpha = -1$: $+0.65$ and $+2.99$ pp), and without the clean reference a distance to the nearest clean centroid detects errors worse than prediction entropy in all six units (-0.020, -0.034 AUROC). Nothing unpaired reconstructs the paired drift signal. The mechanism behind new errors is unknown.
- No established threshold, and no renderer-independent onset. "Threshold" is an inference from four severity levels, not a localised boundary: with no point at 4.5 the crossover is only bounded to (4, 5), and our pre-registered monotonicity prediction failed (P1, 2 of 6 runs). A physically-motivated re-rendering (§8.4) moves the onset to severity 4 while preserving the severity-5 excess, so the supported claim is descriptive — an excess that grows with severity — not a law of the severity axis.
- No generalization to real weather. §7 is observational with insufficient heavy-degradation samples.
- No transfer across families. Coverage of co-occurrence removes the excess on pairs drawn from families the training pools contain, and not on haze+glare, whose families are absent from every lattice pool (§5, P10). A deployment recipe must therefore cover the degradations it expects to meet; we have not shown that it can cover degradations it does not model.

8.3. Data disclosure

The 236 images in FGSC-23's test split were used by three diagnostic analyses on 2026-10-29 (part occlusion, local recovery, restoration probe). They were not used in training or in any number here, but they are no longer an unseen test set and this paper does not call them one; every reported statistic comes from folds inside the training split.

8.4. Threats to validity

Single dataset; 618 training images; two seeds; five severity levels with nothing between 4 and 5.

Two renderers, not one. Re-rendering haze with Koschmieder depth transmission and glare with a specular lobe plus blooming, severity matched on mean transmission and saturated-pixel fraction ($\Delta \leq 0.008$, $\leq$ 3e-8), gives E(5) = +14.6 / +21.8 pp on the frozen baseline and +10.2 / +16.0 pp after retraining (contrast +9.7 / +12.6 pp). The excess belongs to the interaction, not to one clipping model; but the onset moves to severity 4, and saturation matching pins the saturated fraction by construction, so the experiment shows that spatial structure matters at fixed saturation, not that saturation is the cause. Only haze+glare is re-rendered; cloud is unchanged. Remaining caveats: the cloud implementation, one backbone family and one fold in the physical test, and a cluster bootstrap accounting for two seeds per image but not multiplicity over the condition grid. No result here is on an unseen test set (§8.3).

9. Conclusion

Compound weather degradations do to maritime ship classifiers what single degradations do not: at the highest severity the joint effect exceeds the additive prediction by 11.3 pp, exceeds any single degradation, and 41.5% of its errors come from no single degradation. At moderate severities the same conditions are additive; a physical re-rendering keeps the severity-5 excess but moves its onset to 4, so we claim the excess and its growth, not a fixed crossover. Against the union-of-errors baseline the haze+glare excess survives in 6/6 cells (+9.9 pp) while haze+cloud's vanishes (-0.7 pp), and CIFAR-10-C reproduces a smaller excess (+3.6 pp) where the additive baseline reports the opposite sign. Over fifteen calibrated pairs the interaction is additively governed by per-family main effects (R^2 0.42–0.90), whose values are backbone-dependent. A pool drawn from the six families takes the held-out excess from +8.2 to -1.8 pp in 6 of 6 cells, while singly presented families leave it at +7.2 pp; damage matching accounts for ~60% of the gain and co-occurrence for 40%, which generalises to unseen pairs but not unseen families. On CIFAR-10-C coverage removes 8.3 pp too, but single families recover it: there the operative variable is family coverage. Eight single-point interventions failed to beat a combined-augmentation baseline; protocol, predictions and negative results are released together, so the negative half is as checkable as the positive.

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